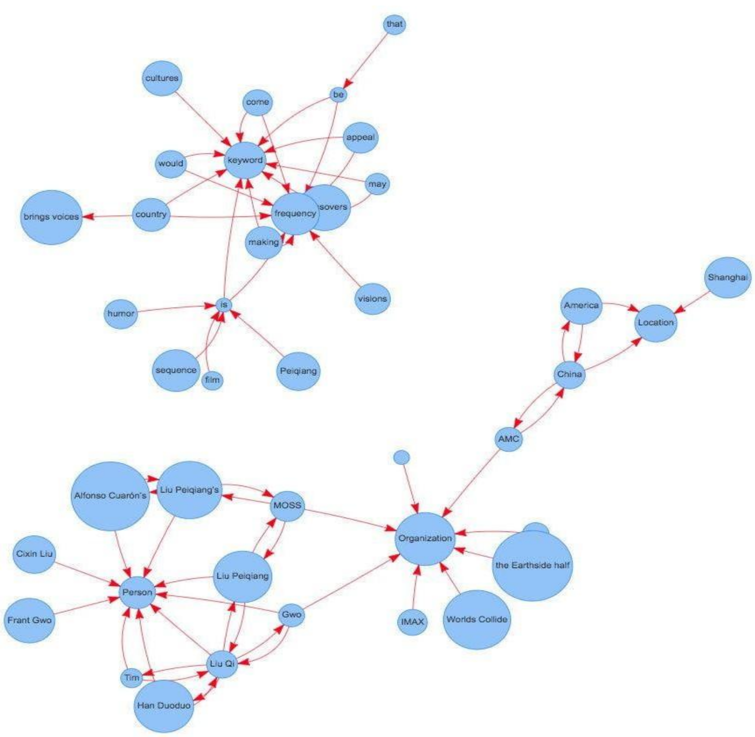
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim:** TextRank for Document Summarization	
Experiment No: 10	**Date:25-04-2026**	**Enrolment No:92301733024**

Aim: TextRank for Document Summarization

IDE: Google Colab

Theory:

TextRank is an algorithm based on PageRank, which often used in keyword extraction and text summarization. In this article, I will help you understand how TextRank works with a keyword extraction example and show the implementation by Python.

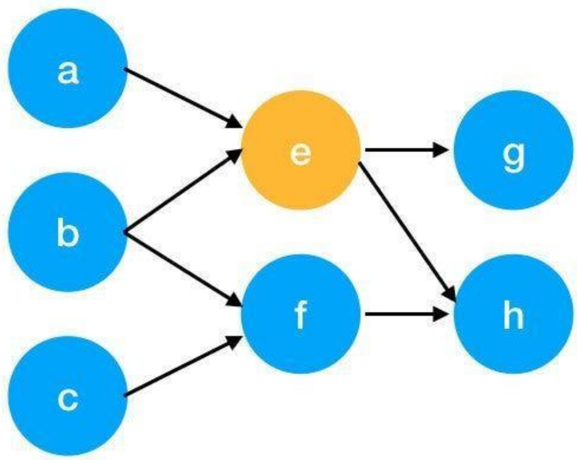


Understand PageRank

There are tons of articles talking about PageRank, so I just give a brief introduction to PageRank. This will help us understand TextRank later because it is based on PageRank. PageRank (PR) is an algorithm used to calculate the weight for web pages. We can take all web pages as a big directed graph. In this graph, a node is a webpage. If webpage A has the link to web page B, it can be represented as a directed edge from A to B. After we construct the whole graph, we can assign weights for web pages by the following formula.

$$S(V_i) = (1-d) + d * \sum_{j \in In(v_i)} \frac{1}{|Out(V_j)|} S(V_j)$$

- S(Vi) - the weight of webpage i
- d - damping factor, in case of no outgoing links
- In(Vi) - inbound links of i, which is a set
- Out(Vj) - outgoing links of j, which is a set
- |Out(Vj)| - the number of outbound links



Here is an example to better understand the notation above. We have a graph to represent how web pages link to each other. Each node represents a webpage, and the arrows represent edges. We want to get the weight of webpage e. We can rewrite the summation part in the above function to a simpler version.

$$In(v_e) = \{a, b\}, j \in \{a, b\}$$

$$\sum_{j \in \{a,b\}} \frac{1}{|Out(V_j)|} S(V_j) = \frac{1}{|Out(V_a)|} S(V_a) + \frac{1}{|Out(V_b)|} S(V_b)$$

$$= \frac{1}{|\{e\}|} S(V_a) + \frac{1}{|\{e, f\}|} S(V_b)$$

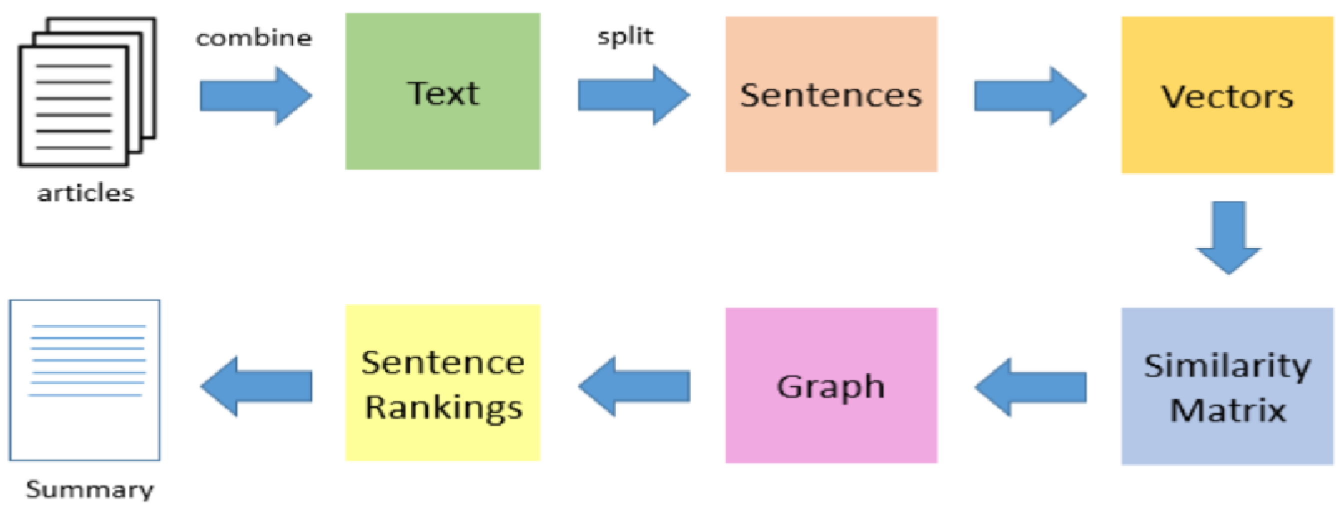
$$= S(V_a) + \frac{1}{2} S(V_b)$$

We can get the weight of webpage e by the following function.

$$S(V_e) = (1-d) + d * \left(S(V_a) + \frac{1}{2} S(V_b) \right)$$

We can see the weight of the webpage e is dependent on the weights of its inbound pages. We need to run this iteration much time to get the final weight. In the initialization, the importance of each webpage is 1.

TextRank for document Summarization



TextRank works in the following steps:
1. Tokenize documents into sentences.
2. Preprocess each sentence in the document.
3. Count key phrases and normalize them or produce TFIDF Matrix, you can also use any kind of vectorization such as spacy vectors.
4. Calculate the Jaccard Similarity between sentences and key phrases.
5. Rank the sentences with higher significance.

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Pre Lab Exercise:

1. How keyword extraction is similar to document summarization process?
    - Both use the TextRank algorithm based on PageRank.
    - Both convert text into a graph structure
    - Both rank elements using importance scores**.**
    - Both are unsupervised methods
    - Both aim to extract the most important information from the text.

2. How keyword extraction is different to document summarization process?

    Key word extraction focuses on identifying important words or phrases from a text, while document summarization focuses on selecting key sentences to create a shorter version. Keyword extraction gives a list of terms, whereas summarization provides a meaningful paragraph.

3. Limitation of TextRank as a summarizer
    - Does not understand deep meaning or context
    - May miss important sentences
    - Depends on sentence similarity
    - Can produce incomplete summaries
    - May include irrelevant or less important sentences

Program (Code):

```
# Install libraries
!pip install pytextrank spacy
!python -m spacy download en_core_web_sm
# Import libraries
import spacy
import pytextrank
# Load model
nlp = spacy.load("en_core_web_sm")
nlp.add_pipe("textrank")
# Input text (portfolio)
text = """
My self Isha Savaliya and I am a B.Tech ICT student at Marwadi University.
I completed my 12th from Shree GK Dholakiya Scool.
I am interested in Web Developmenmt and want to build my career in this field.
My goal is to get a good job and earn money.
I love to watch series and movies.
I like to play games.
Sometimes I feel angry and frustrated but I try to improve myself.
"""
# Apply TextRank
```

```
doc = nlp(text)

# Print top sentences (summary)
print("Top Ranked Sentences:\n")
for sent in doc._.textrank.summary(limit_phrases=10, limit_sentences=5):
    print(sent)
```

```
Top Ranked Sentences:

My self Isha Savaliya and I am a B.Tech ICT student at Marwadi University.

I completed my 12th from Shree GK Dholakiya Scool.

I love to watch series and movies.

I am interested in Web Developmenmt and want to build my career in this field.

I like to play games.
```

Post Lab Exercise:

1. Write about “your ownself” (in not more than 500 words You know better about you, rather than ChatGPT!!). Generate the summary of your portfolio in
   a. 50% of the size of your portfolio
   b. 25% of the size of your portfolio

Comment over the summary obtained. Which sentence you think should be there in your summary, but was not spotted by textrank? Rate the overall summary in the scale of 1-5 (with 1 as least and 5 as highest rating). Paste the code, your portfolio, output and your analysis.

```
!pip install nltk scikit-learn networkx

import nltk nltk.download('punkt')
nltk.download('punkt_tab')

import numpy as np import
networkx as nx
from nltk.tokenize import sent_tokenize
from sklearn.feature_extraction.text import TfidfVectorizer from
sklearn.metrics.pairwise import cosine_similarity

text = """
I am a B.Tech student in Information and Communication Technology, currently in Semester 6. I am passionate about Artificial Intelligence, especially in real-world applications like education and healthcare.
I am currently working on an AI-based Answer Sheet Evaluation System that uses OCR and NLP techniques.
I have knowledge of Python, Java, PHP, and Dart.
I have worked with Laravel, Flutter, and MySQL.
I enjoy building practical projects and solving real-world problems. My goal is to
build intelligent systems that improve efficiency.
    """

sentences = sent_tokenize(text)

vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(sentences) similarity_matrix =

cosine_similarity(tfidf_matrix, tfidf_matrix)

nx_graph = nx.from_numpy_array(similarity_matrix) scores =
nx.pagerank(nx_graph)

ranked_sentences = sorted(((scores[i], s) for i, s in enumerate(sentences)), reverse=True) print("\nTop 50%

Summary:\n")
top_n = int(len(sentences) * 0.5)

for i in range(top_n):
        print("-", ranked_sentences[i][1])
```

```
print("\nTop 25% Summary:\n")
top_n = max(1, int(len(sentences) * 0.25))

for i in range(top_n):
    print("-", ranked_sentences[i][1])
```

```
...
  Top 50% Summary:

  - I am passionate about Artificial Intelligence, especially in real-world applications like education and healthcare.
  -
  I am a B.Tech student in Information and Communication Technology, currently in Semester 6.
  - I am currently working on an AI-based Answer Sheet Evaluation System that uses OCR and NLP techniques.

   Top 25% Summary:

  - I am passionate about Artificial Intelligence, especially in real-world applications like education and healthcare.
```